

Daily Algebra Drill #2

Section	Topic	Questions	Target time
A	Rapid Recognition	10	2 min 30 sec
B	Manipulation Drills	10	8 min
C	Substitution & Structure	8	10 min
D	Challenge	5	15 min

New toolkit today: Vieta's formulas for cubics · Nested surds · Telescoping products · Lagrange/Brahmagupta identity · Factor theorem · Cyclic sums under constraints.

Attempt each question before checking answers. Time yourself. No calculators.

Section A Rapid Recognition

(≤15 s each)

Factorise, simplify, or evaluate instantly. No expansion.

- Factorise completely: $x^3 - 27$.
- Without expanding, evaluate $(\sqrt{5} + \sqrt{3})(\sqrt{5} - \sqrt{3})$.
- Without a calculator, evaluate $999^2 - 1$.
- Given $a - b = 4$ and $ab = 5$, find $a^2 + b^2$.
- Expand and simplify: $(x + y)^3 - x^3 - y^3$.
- Factorise completely: $3x^2 - 75$.
- Rationalise the denominator: $\frac{1}{\sqrt{3} + \sqrt{2}}$.
- Factorise: $x^2(x - 1) - (x - 1)$.
- The roots of $x^2 - 5x + 3 = 0$ are p and q . Find $p^2 + q^2$.
- Simplify: $\frac{2^{n+3} - 2^n}{2^n}$.

Section B Manipulation Drills

(≤50 s each)

Fractions, rearranging, hidden factorisations. Elegance over brute force.

- Simplify: $\frac{x}{x^2 - 1} + \frac{1}{x - 1}$.
- Factorise: $x^4 - x^2 - 12$.
- Make r the subject of $S = \frac{a}{1 - r}$.

4. Simplify: $\left(1 + \frac{1}{x}\right)\left(1 - \frac{1}{x+1}\right)$.
5. Given $x + y = 3$ and $xy = 1$, find $x^3 + y^3$.
6. Simplify: $\frac{(a+b)^2 - (a-b)^2}{4ab}$.
7. Factorise: $x^3 + 3x^2 - x - 3$.
8. Solve: $\frac{x-1}{x+2} = \frac{x+3}{x+8}$.
9. Simplify: $\frac{\sqrt{a}-\sqrt{b}}{\sqrt{a}+\sqrt{b}} + \frac{\sqrt{a}+\sqrt{b}}{\sqrt{a}-\sqrt{b}}$.
10. The cubic $x^3 - 6x^2 + 11x - 6 = 0$ has roots α, β, γ . Without solving it, find $\alpha^2 + \beta^2 + \gamma^2$ and $\alpha\beta\gamma$.

Section C Substitution & Structure

(≤90 s each)

Find structure before computing. Substitution is the key, not expansion.

1. Solve: $x^4 - 13x^2 + 36 = 0$.
2. Solve: $(x+1)(x+2)(x+3)(x+4) = 24$.
3. Simplify: $\sqrt{6 + 2\sqrt{5}}$.
4. Given $a + b + c = 1$ and $a^2 + b^2 + c^2 = 1$, find the value of $ab + bc + ca$ and hence evaluate $a^3 + b^3 + c^3$.
5. Solve: $(x-1)(x-3)(x-5)(x-7) + 15 = 0$.
6. If $x = 2 + \sqrt{3}$, find $x^2 + \frac{1}{x^2}$ without expanding x^2 directly.
7. Given $\frac{a}{b} + \frac{b}{a} = 3$, find $\frac{a^2}{b^2} + \frac{b^2}{a^2}$.
8. If $x + y + z = 0$, simplify $\frac{(x^2 + y^2 + z^2)^2}{x^4 + y^4 + z^4}$.

Section D Challenge

(TMUA / SMC difficulty)

Each has a solution under five lines. Find the insight, not the computation.

1. Evaluate:

$$\frac{1}{1 \cdot 2 \cdot 3} + \frac{1}{2 \cdot 3 \cdot 4} + \frac{1}{3 \cdot 4 \cdot 5} + \cdots + \frac{1}{8 \cdot 9 \cdot 10}.$$

2. The roots of $x^2 + px + q = 0$ are r and s . Find, in terms of p and q , the monic quadratic whose roots are $r^2 + s$ and $s^2 + r$.
3. The equation $\frac{x}{y+z} = \frac{y}{x+z} = \frac{z}{x+y} = k$ holds for real x, y, z with $x + y + z \neq 0$. Find all possible values of k .
4. Prove the **Brahmagupta–Fibonacci identity**:

$$(a^2 + b^2)(c^2 + d^2) = (ac + bd)^2 + (ad - bc)^2.$$

Hence express 85×65 as a sum of two perfect squares in two different ways.

5. Find all integers $n \geq 1$ for which $n^2 + n + 1$ divides $n^3 - 1$.

“An elegant solution is not luck — it is the reward for seeing the structure that was always there.”

other prep to look at today:
Daily Integral for Calculus and Logic